

8. FOURTH WEEK OF EMBRYONIC DEVELOPMENT

The fourth week of embryonic development is marked by several phenomena:

- The delimitation of the embryo which takes its final form
- The formation of the organ primordia (becoming leaflets, each leaflet evolves on its own and gives rise to its main derivatives).

1. EMBRYO DELIMITATION (PLICATURE)

It begins at the end of the 3rd week and extends throughout the 4th week. This is a ventral coiling of the embryo to differentiate itself from its annexes. The delimitation can be described on both transverse and median longitudinal sections.

The significant development of the amniotic cavity will cause it to turn under the embryo to completely surround it.

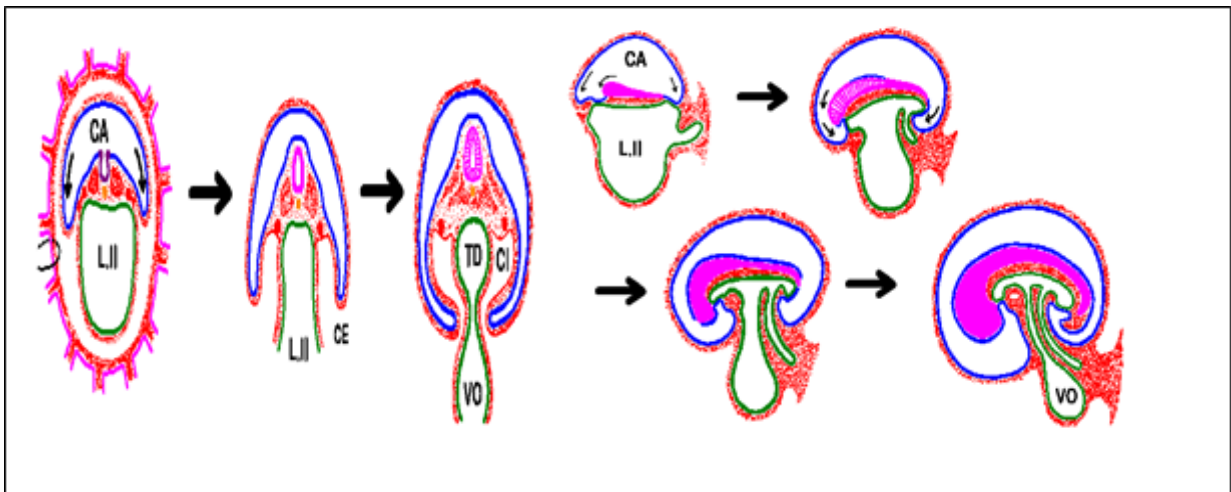


Figure 1. Transverse and longitudinal delimitation of the embryo

The result will be that:

- the embryo will be completely surrounded by ectoderm;
- the cranial and caudal extremities will undergo a very significant folding, resulting in a complete reversal of the pharyngeal and cloacal membranes.
- the yolk sac will be constricted, forming:
 - **an upper part**, in the form of a longitudinal tube, the primitive digestive tract, limited at its ends by the pharyngeal and cloacal membranes;

- **a lower part**, forms a vesicle which retains the name vitelline (or umbilical) vesicle; between the two, the vitelline (or umbilical) duct;
- The embryonic stalk is pushed under the embryo; it comes into a ventral position and thus the umbilical cord is formed, limited by the amniotic epithelium and containing the yolk sac and the allantois.
- The cardiac primordium is also pushed into a ventral position and will end up inside the embryo, towards the anterior 1/3.

The embryo, delimited by the ectoderm, is thus bathed in the amniotic cavity, which is now increasing in volume, and the external coelom is compressed between the amniotic membrane and the chorionic membrane. The embryo is connected to the chorion (the precursor of the placenta) by the umbilical cord, through the amniotic cavity.

2. EVOLUTION OF THE THREE FUNDAMENTAL LAYERS (beginning of organogenesis)

2.1. Evolution of the ectoderm

Neurulation is completed during the 4th week. The neural groove fuses at points, beginning near the middle of the tube. At the fusion points, the neural crests become isolated and segment into a series of units: the ganglionic primordia (future spinal ganglia).

The mechanism extends and the neural tube ends with two openings: the anterior and posterior neuropores (the anterior neuropore closes at 25 or 26 days and the posterior neuropore at 28 days).

The neural tube will give rise to the central nervous system. It is a hollow tube (future spinal cord) with a wider cranial end (future brain) which rotates 180° and folds under the ventral side of the embryo.

Before the end of the 4th week, the cephalic region shows 3 successive dilations, the cerebral vesicles, which are from front to back:

- **The proencephalon:** forebrain, is divided into two parts: telencephalon (cerebral hemispheres) and diencephalon (optic pathways, hypothalamus, pituitary gland).
- **The mesencephalon:** midbrain, auditory reflexes and visual input
- **The rhombencephalon:** hindbrain, is divided into two parts: metencephalon (cerebellum) and myelencephalon (medulla oblongata).

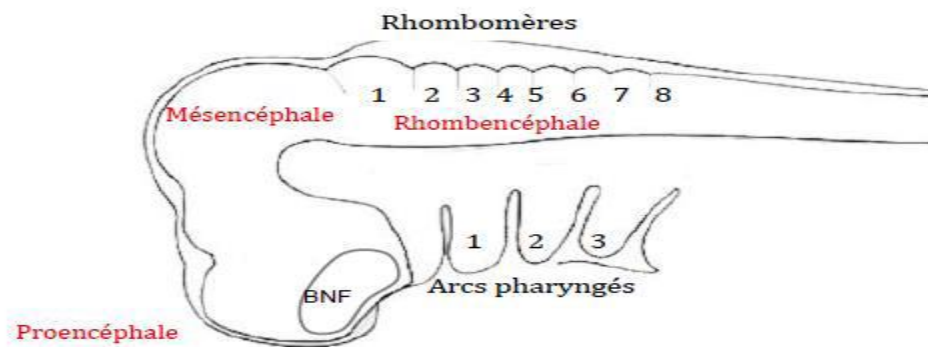


Figure 2. Neural tube evolution

2.2. Evolution of the mesoderm

• Para-axial mesoderm

The first somites appear on the 19th day at the anterior part and the last ones, at the caudal end, on the 40th day.

There will be a total of 42 to 44 pairs of somites distributed as follows: 3 to 4 occipital pairs (transient and poorly individualized), 8 cervical pairs, 12 dorsal pairs, 5 lumbar pairs, 5 sacral pairs and 8 to 12 coccygeal pairs.

The somites allow for the morphological dating of the embryo:

- 21 days: 4 to 7 pairs of somites
- 23 days: 10 to 13 pairs of somites
- 25 days: 17 to 20 pairs of somites
- 28 days: 26 to 28 pairs of somites
- 30 days: 30 to 32 pairs of somites
- 40 days: 42 to 44 pairs of somites.

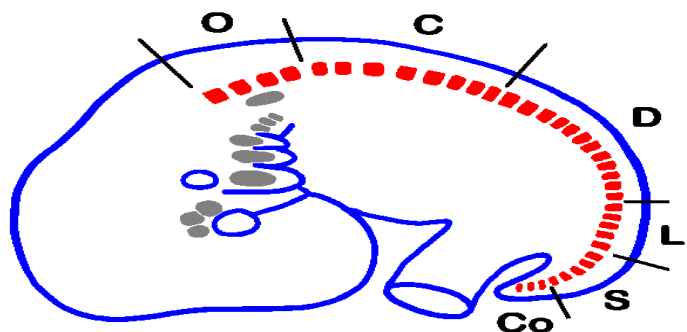


Figure 3. Somite formation

The somites quickly divide into two parts:

- **The sclerotome** : The internal structure will give rise to the axial skeleton, under the inductive influence of the notochord and neural tube. The metameric condensation of the

sclerotome around the notochord and neural tube will give rise to the formation of the vertebrae.

- **The dermomyotome:** the two parts quickly separate by cleavage into dermatome and myotome. The dermatome itself will diffuse With elements originating from the intra-embryonic somatopleure, it will give rise to the dermis. As for the myotomes, they will undergo strictly myogenic differentiation, and thus produce the majority of the body's muscles, including those of the limbs.

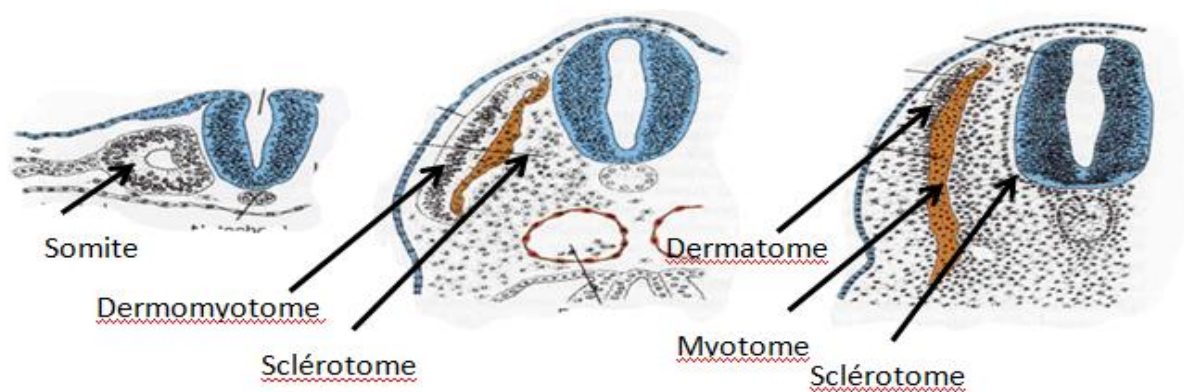


Figure 4. Somite development

- **Intermediate mesoderm**

The intermediate mesoderm also segments in turn from the 2nd occipital metamere to the 4th lumbar metamere to form small cellular clusters, the nephrotomes.

This metamerization shows three regions from front to back:

- **The pronephros** extends from the 2nd pair of occipital somites to the 4th pair of the 4th lumbar metamere. From day 19 to day 25. But the pronephros immediately begins to involute and disappears on day 30.
- **The mesonephros** extends from the 5th cervical to the 4th lumbar nerve by day 20 and is complete by day 40, but half of the nephrotomes involute; the rest will form the urinary tract and the male genital tract.
- **The metanephros** It extends from the 5th lumbar somites to the 5th sacral somites. There is no segmentation into nephrotomes, and the metanephros remains as a solid band called the metanephrogenic blastema which is the rough draft of the definitive kidney.

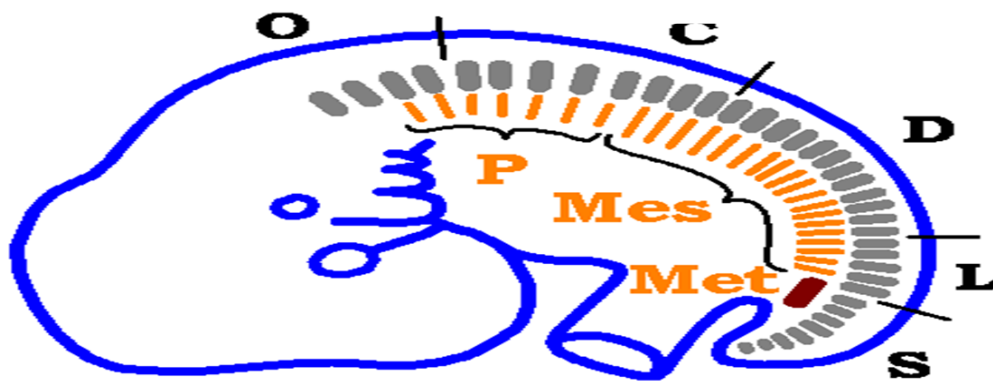


Figure 5. Development of the intermediate mesoderm

- **Lateral mesoderm**

- **The intra-embryonic somatopleure** : The parietal layer will line the wall of the intra-embryonic coelom and, together with the ectoderm, will form the lateral and ventral walls of the embryo.
- **The intra-embryonic splanchnopleure** : The visceral layer, surrounding the organs, will form, together with the endoderm, the wall of the digestive tract.
- Between the two, the intra-embryonic coelom is the precursor of the future pericardial-pleural-peritoneal cavity.

2.3. Evolution of the endoderm

By the end of the 4th week, the primitive digestive tract comprises:

- **Anterior primitive intestine**

It includes the oral cavity, formed during the resorption of the pharyngeal membrane, the esophagus, and ends in a small dilation, the primitive stomach. It gives rise to a ventral bud which forms the respiratory system.

- **primitive middle intestine**

It initially gives rise to hepatic, cystic, and pancreatic buds. It corresponds to the primitive intestinal loop that gives rise to the small intestine and half of the large intestine.

- **primitive posterior intestine**

It is the origin of the second half of the large intestine. Its end empties into the cloaca (urogenital sinus anteriorly and rectum posteriorly).

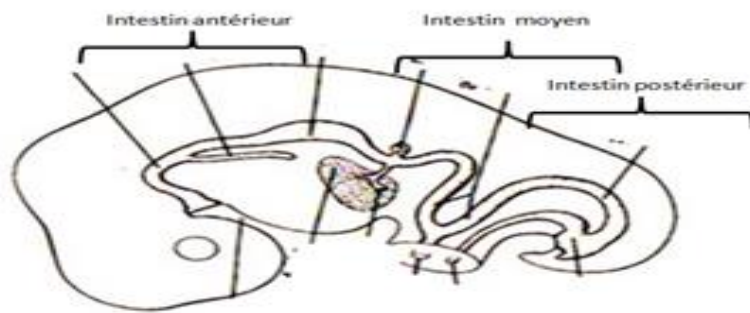


Figure6. Development of the primitive digestive tract

7.2.4. Evolution of the mesenchyme

By the end of the 4th week, circulation is functional and ensures the nutrition of growing embryonic tissues. In the extra-embryonic mesenchyme and intra-embryonic, numerous vascular primordia develop.

- **At the level of the chorionic villi, the chorionic sphere, and the pedicle:**

The umbilical vessels use the umbilical cord and connect to the intra-embryonic vascularization.

- **Around the yolk sac:**

The cardiogenic zone moves to a ventral position and becomes intra-embryonic, forming the primitive, unpaired, and median cardiac tube. The first heartbeats appear on the 24th day.

Blood circulation will be directed and functional from 28-29 days.

At the end of the 4th week, the embryo is delimited and takes its final shape.

The chorionic sphere is 30 mm in diameter and the embryo is 5 mm long.

Blood circulation was established and the beginnings of many organs were formed.

Due to the importance of the phenomena, the 4th week is a very critical phase from a teratological point of view.